GCSE GEOGRAPHY

Paper 3: Geographical applications

Topic 7: Geographical applications (issue evaluation and fieldwork)

Practice questions in the style of AQA GCSE Geography (8035). Every question is original; the layout, command words, mark allocations (including levels-based 6- and 9-mark questions with SPaG) and mark scheme follow the real exam format. Figures and data are illustrative and simplified for practice; case-study facts in the mark scheme are summarised, so check them against your own case-study notes.

Time allowed	About 90 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a black pen, a pencil, a ruler, a calculator and (for some questions) a protractor.
Instructions	• Use black ink or black ball-point pen. Use pencil only for drawing. • Answer **all** questions. • You must answer the questions in the spaces provided. • Do all rough work in this book. Cross through any work you do not want to be marked. • Questions that carry a **SPaG** note are also marked for spelling, punctuation, grammar and the use of specialist terminology.
Information	• The maximum mark for this paper is **87**. • The marks for questions are shown in brackets. • Questions that start with the command word *Explain* need developed points (a point plus a reason or consequence). • 6- and 9-mark questions are marked by **levels**: aim to give a detailed, balanced and well-linked answer, using place-specific detail.

Name: _____ Date: _____ Mark: _____ / 87

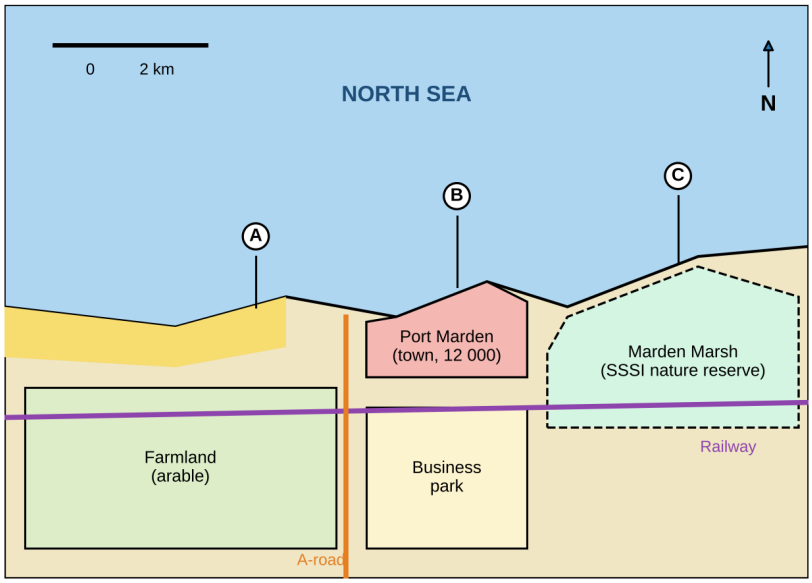
Section A: Issue evaluation

Answer all questions in this section.

0 1 **Section A: Issue evaluation**. This section uses **Resources 1 to 5**. The issue is: *How should the town of Port Marden manage the risk of coastal flooding?* Port Marden is a coastal town of 12 000 people on the east coast of England. Its flood defences are failing, and the council must choose between three options: a sea wall (Option 1), managed retreat (Option 2) or a tidal barrier (Option 3).

0 1 . 1 Study **Resource 1**. In which direction is Marden Marsh from Port Marden? **[1 mark]**

Resource 1: Map of Port Marden and surrounding area



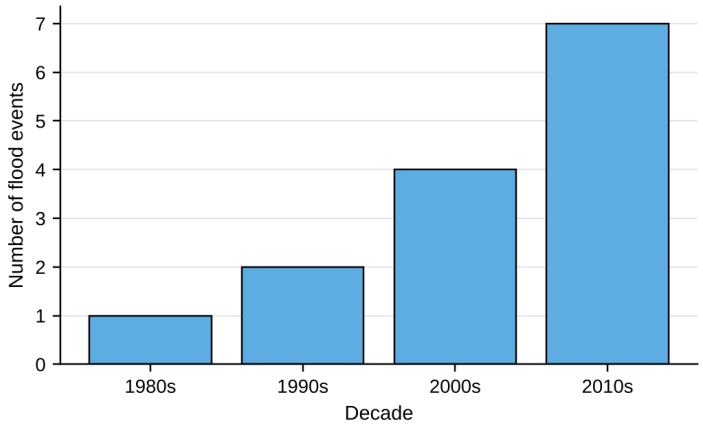
0 1 . 2 Using the scale on Resource 1, measure the straight-line distance from the centre of Port Marden to the centre of Marden Marsh. **[2 marks]**

Answer = _____ km

0 1 . 3 Name the land uses at points **A** and **C** on Resource 1. **[2 marks]**

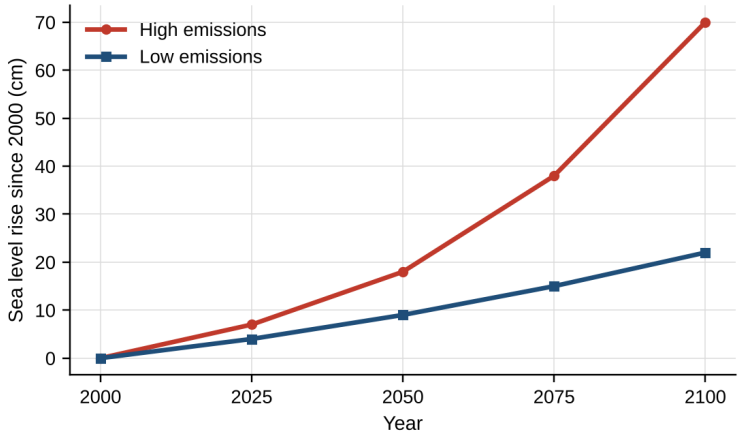
0 1 . 4 Study **Resource 2**. Describe the trend in the number of flood events in Port Marden. **[2 marks]**

Resource 2: Flood events in Port Marden



0 1 . 5 Study **Resource 3**. Calculate the difference in sea level rise between the high and low emissions scenarios in 2100. **[2 marks]**

Resource 3: Predicted sea level rise



Answer = _____ cm

0 1 . 6 Suggest **two** reasons why the risk of flooding in Port Marden is likely to increase in future. **[4 marks]**

0 1 . 7 Study **Resource 4**. Calculate the cost per home protected for the sea wall (Option 1). Give your answer to the nearest pound. **[2 marks]**

Resource 4: Comparison of the three options

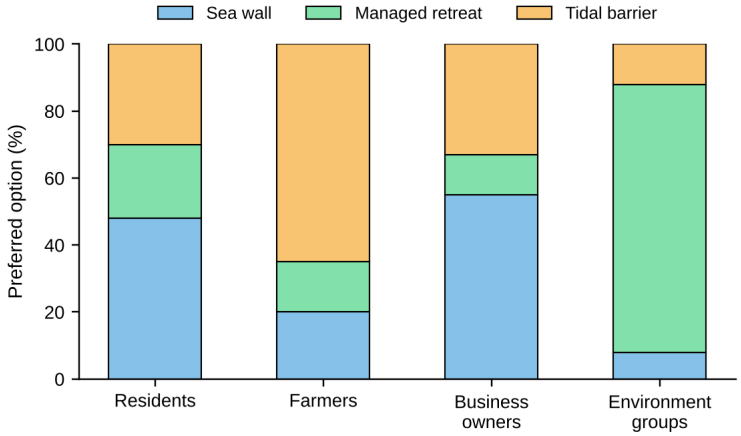
	Option 1 Sea wall	**Option 2 Managed retreat**	**Option 3 Tidal barrier**
Cost	£38 million	£9 million	£85 million
Homes protected	1 200	0 (140 homes lost)	1 200
Other benefits	Protects the road, railway and business park	Creates 80 hectares of salt marsh habitat	Generates electricity for 20 000 homes
Lifespan	60 years	Not applicable	100 years
Main problems	Visual impact; erosion increases further along the coast	Loss of 140 homes and some farmland; compensation costs	Very high cost; affects marsh habitats; long construction time

Answer = _____ £

0 1 . 8 Which option is **cheapest** and which is **most expensive**? **[2 marks]**

0 1 . 9 Study **Resource 5**. Which group supports managed retreat the most? Give the figure. **[2 marks]**

Resource 5: Survey of preferred option by group



0 1 . 10 Explain **one** advantage and **one** disadvantage of the sea wall (Option 1) for Port Marden. **[4 marks]**

0 1 . 11 Using **all the resources** and your own knowledge, evaluate which option should be chosen to manage the flood risk at Port Marden. You should make a justified recommendation and use evidence from the resources to support it. **[9 marks]**

0 2 **Section B: Fieldwork**. A group of students studied how a river changes downstream. They collected data at ten sites along the River Marden from 1 km to 10 km from its source. **Resource 6** shows their velocity results.

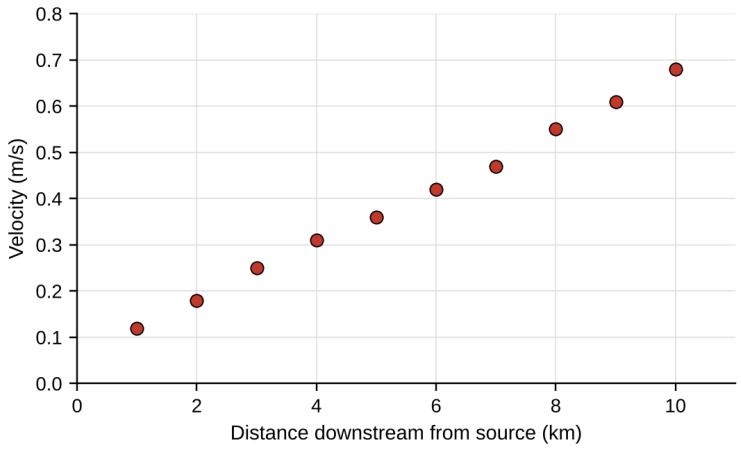
0 2 . 1 Give **one** geographical question the students could investigate. **[1 mark]**

0 2 . 2 Describe a method the students could use to measure the **velocity** of the river at each site. **[3 marks]**

0 2 . 3 Suggest **one** risk when carrying out river fieldwork and explain how it could be reduced. **[2 marks]**

0 2 . 4 Use Resource 6 to describe the relationship between distance from the source and velocity. **[3 marks]**

Resource 6: Velocity results



0 2 . 5 The velocity results (m/s) for the ten sites are: 0.12, 0.18, 0.25, 0.31, 0.36, 0.42, 0.47, 0.55, 0.61, 0.68. Calculate the **mean** and the **median** velocity. **[3 marks]**

Answer = _____ m/s

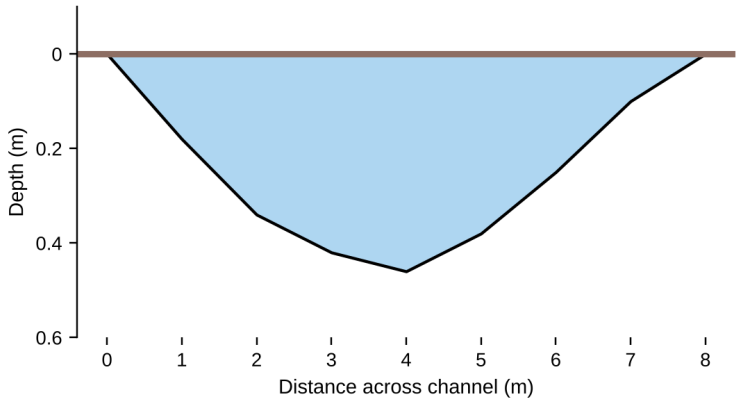
0 2 . 6 Calculate the range of the velocity results. **[1 mark]**

Answer = _____ m/s

0 2 . 7 Suggest **one** reason why velocity increases downstream, even though the gradient decreases. **[2 marks]**

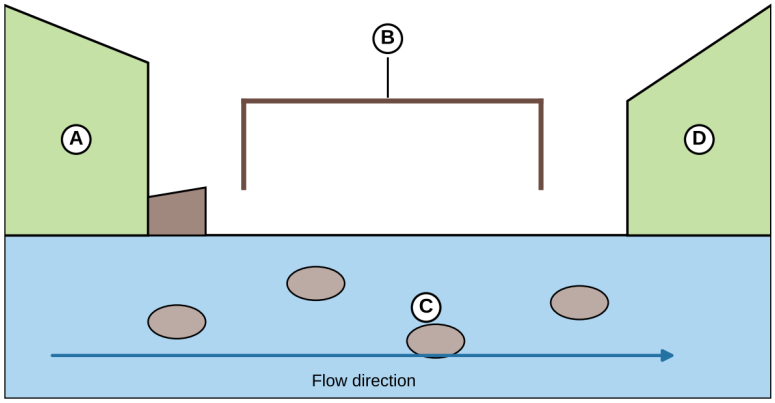
0 2 . 8 **Resource 7** shows a cross-section of the channel at one site. Calculate the width and the maximum depth. Describe the shape. **[3 marks]**

Resource 7: Channel cross-section



0 2 . 9 Resource 8 is a field sketch of one of the sites. Name features **B** and **C**, and suggest how the sketch could help the students. **[3 marks]**

Resource 8: Field sketch of the site



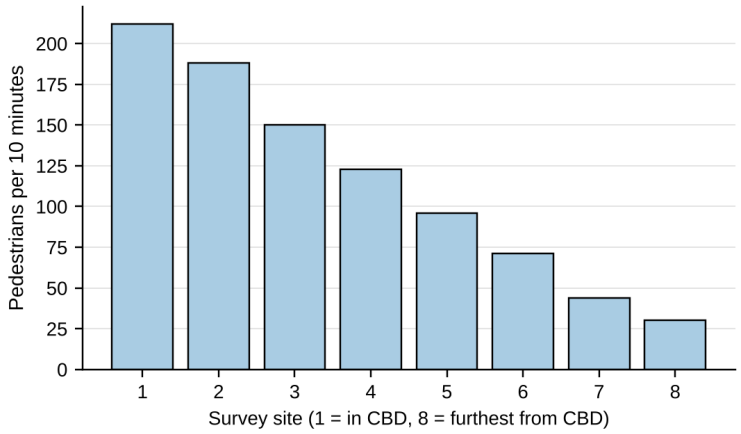
0 2 . 10 Evaluate the reliability of the students' results and suggest **two** ways in which they could improve their investigation. **[6 marks]**

0 3 A second group of students investigated **how environmental quality and pedestrian flow change with distance from the CBD** in a town. **Resource 9** shows their pedestrian counts.

0 3 . 1 Describe how the students could carry out a pedestrian count at one site. **[3 marks]**

0 3 . 2 Describe the pattern shown in Resource 9. **[3 marks]**

Resource 9: Pedestrian counts



0 3 . 3 Calculate the mean number of pedestrians per site. **[2 marks]**

Answer = _____

0 3 . 4 The students ranked each site by pedestrian count (Rank A) and by environmental quality score (Rank B). **Table 1** shows their ranks. Complete the d^2 column and calculate Σd^2. **[3 marks]**

Table 1

Site	Rank A	Rank B	d	d^2
1	1	2		
2	2	1		
3	3	4		
4	4	3		
5	5	6		
6	6	5		
7	7	8		
8	8	7		

0 3 . 5 Use the formula $Rs = 1 - (6\Sigma d^2) \div (n(n^2 - 1))$ with $n = 8$ to calculate the Spearman's rank correlation coefficient. **[3 marks]**

Answer = _____

0 3 . 6 What does the Rs value in the previous question tell you about the relationship between pedestrian flow and environmental quality? **[2 marks]**

0 3 . 7 Suggest **two** ways in which the students could present their environmental quality results. **[2 marks]**

0 3 . 8 Explain why the students' conclusion about distance from the CBD may **not be valid** for all towns. **[4 marks]**

0 3 . 9 **'Primary data collected by fieldwork is more useful than secondary data.'** To what extent do you agree? Use your own fieldwork experience. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 7: Geographical applications (issue evaluation and fieldwork)

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *geography*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	east	Allow 'E' or 'east'. Do not accept 'north-east'.	1	AO3 3.3.1
01.2	measurement 4.5–4.8 units (46–50 mm on printed map) ; 3.0–3.3 km ;	Allow 2.9–3.4 km. Scale: 3 units on the map = 2 km.	2	AO4 3.3.1
01.3	A: beach / sand / coastline near farmland ; C: salt marsh / nature reserve (SSSI) ;	Accept sensible descriptions: A = beach / sandy coast; B = town; C = marsh.	2	AO3 3.3.1
01.4	increase / rise in number of floods ; from 1 in the 1980s to 7 in the 2010s / increases each decade ;		2	AO3 3.3.1
01.5	70 − 22 ; 48 (cm) ;	Allow 46–50.	2	AO4 3.3.1
01.6	sea level rising because of global warming (melting ice sheets, thermal expansion) ; rising to 22–70 cm by 2100 (Resource 3), so waves reach further / existing defences are too low ; storm surges / stormier weather more frequent with climate change ; number of flood events has increased each decade (Resource 2) ; land may be sinking / existing defences failing ;	Two reasons, each with 1 mark for the reason and 1 for use of resources / development.	4	AO2/AO3 3.3.1
01.7	38 000 000 ÷ 1 200 ; £31 667 ;	Allow £31 600–31 700. Correct answer with no working scores 2.	2	AO4 3.3.1
01.8	cheapest: Option 2 (managed retreat) £9 million ; most expensive: Option 3 (tidal barrier) £85 million ;		2	AO3 3.3.1
01.9	environment groups ; 80% (allow 78–82%) ;	Read from the middle (green) section: 88 − 8 = 80.	2	AO3 3.3.1
01.10	advantage: protects 1 200 homes and the road, railway and business park ; so jobs and the local economy are not lost / people feel safe ; disadvantage: expensive (£38 million) ; may increase erosion along the coast / ugly / does not last as long as the barrier (60 years) ;	Each point developed using Resource 4.	4	AO3 3.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.11	**Level 3 (7–9 marks)**: Detailed, accurate knowledge and understanding of the topic with a well-developed, balanced and logical argument. A clear, supported **evaluation/judgement** that is based on evidence (and specific place detail or the resources where asked). Any conclusion is fully justified. **Level 2 (4–6 marks)**: Some accurate knowledge and understanding. Some development of ideas with evidence; the judgement/evaluation is partial, unbalanced or not fully supported. **Level 1 (1–3 marks)**: Limited knowledge and understanding. Simple statements with little development; a judgement, if any, is unsupported. **0 marks**: No relevant content.	Marks for the quality of the evaluation, not for the option chosen.	9	AO1/AO2/AO3 3.3.1
	Indicative content (credit any other valid geography, and place-specific detail): • Option 1 (sea wall): protects 1 200 homes plus infrastructure ; cost per home about £31 667 ; popular with residents (48%) and business owners (55%) ; but visual impact / erosion elsewhere ; • Option 2 (managed retreat): cheapest (£9 million), creates 80 ha habitat ; supported by environment groups (80%) ; but 140 homes lost and compensation ; opposed by farmers / residents ; • Option 3 (tidal barrier): protects the town for 100 years, powers 20 000 homes ; supported by farmers (65%) ; but very expensive (£85 million) and affects marsh habitats ; • Links to resources: floods increasing 1 → 7 events per decade ; sea level rise up to 70 cm by 2100 (high scenario) – long-term protection needed ; • Balanced conclusion: a clear recommendation with justification, e.g. 'Option 1 offers good value but Option 3 is the best long-term solution' ; recognises that some people will lose out ;			
02.1	e.g. 'How does river velocity change downstream?' / 'How does the channel shape change with distance from the source?'	Must be a question relating to a river and a change with distance.	1	AO3 3.3.2
02.2	use a flow meter / or float (orange / dog biscuit) ; measure a set distance (e.g. 10 m) along the river and time how long the float takes to travel between the points ; repeat several times and calculate the mean / speed = distance ÷ time ; measure in the middle of the channel / at the same place at each site ;		3	AO3 3.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	risk: slipping / drowning in deep or fast water / cold water / contaminated water ; reduce: wear wellingtons or a life jacket, work in pairs with a leader, avoid deep or flooded rivers, check weather first, wash hands after ;		2	AO3 3.3.2
02.4	positive relationship / velocity increases with distance downstream ; from 0.12 m/s at 1 km to 0.68 m/s at 10 km ; fairly steady / linear increase (no anomalies) ;		3	AO3 3.3.2
02.5	sum = 3.95 ; mean = 0.395 (allow 0.40) ; median = (0.36 + 0.42) ÷ 2 ; 0.39 (m/s) ;	Mean 0.395; median 0.39. Allow 0.4 for mean.	3	AO4 3.3.2
02.6	0.56 (m/s)	0.68 − 0.12 = 0.56.	1	AO4 3.3.2
02.7	channel is larger / bigger cross-section, so less friction with the bed and banks ; channel is smoother (smaller, rounded bed material) ; discharge increases as more tributaries join ;		2	AO2 3.3.2
02.8	width 8 m ; maximum depth 0.46 m (allow 0.4–0.5) ; asymmetrical / slightly deeper to one side ; shallow at the edges, deepest around 4 m ;		3	AO3/AO4 3.3.2
02.9	B: bridge ; C: boulders / bed load (large stones on the river bed) ; sketch records features / how the site looked, to help explain results, e.g. coarse bed load may slow the flow ;		3	AO3 3.3.2
02.10	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO3 3.3.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
	Indicative content (credit any other valid geography, and place-specific detail): • only one measurement of velocity per site, so results may be unreliable ; repeat and take the mean ; • only ten sites from a single river – results may not apply to all rivers ; study another river / more sites ; • different group members measure ; human error with floats, using a flow meter is more accurate ; • river conditions vary with the weather: collect data on the same day or after a similar time of rainfall ; • positives: clear pattern / no anomalies, a large range of distances ; • judgement / conclusion about confidence in the results ;			
03.1	stand at the same location for a fixed time (e.g. 10 minutes) ; use a tally chart to record every pedestrian passing a line ; repeat at different times of the day and take the mean ; work in pairs / count both directions ;		3	AO3 3.3.3
03.2	pedestrian numbers fall with distance from the CBD ; from 212 at site 1 to 30 at site 8 ; biggest fall between sites 1 and 3 / consistent decrease ;		3	AO3 3.3.3
03.3	212 + 188 + 150 + 123 + 96 + 71 + 44 + 30 = 914 ; 914 ÷ 8 = 114.25 / 114 ;	Allow 114.	2	AO4 3.3.3
03.4	d = 1 for every site ; $d^2 = 1$ for each site ; $\Sigma d^2 = 8$;	d = 1 or −1; $d^2 = 1$; $\Sigma d^2 = 8$.	3	AO4 3.3.3
03.5	6 × 8 = 48 ; 8 × 63 = 504 ; 48 ÷ 504 = 0.095 ; 1 − 0.095 = 0.905 (allow 0.90 / 0.91) ;	Rs = +0.905.	3	AO4 3.3.3
03.6	strong positive correlation / very close to +1 ; higher pedestrian counts are associated with higher environmental quality (sites ranked similarly) ;	For n = 8, the critical value at 95% is 0.74; accept a reference to this.	2	AO3 3.3.3
03.7	bar chart / compound bar ; radial (radar) graph ; map with proportional symbols / colour shading ; annotated photographs / field sketch ;		2	AO3 3.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.8	only one town studied, with its own layout / history ; environmental quality may be affected by other factors e.g. traffic, wealth, age of housing ; small sample of eight sites ; counts taken only at one time of day ; subjective environmental scoring varies between people ; weather conditions, events may affect pedestrian flow ;		4	AO3 3.3.3
03.9	**Level 3 (5–6 marks)**: Detailed, accurate and relevant knowledge and understanding of the topic. Clear, well-developed and logically structured explanation/analysis with specific links made between ideas (and a use of place detail or the resources where asked). **Level 2 (3–4 marks)**: Some relevant and accurate knowledge and understanding. Some explanation/analysis, but developed unevenly or with limited links. Some use of detail or the resources. **Level 1 (1–2 marks)**: Basic, general knowledge with limited accuracy or relevance. Simple statements; little or no explanation or use of detail. **0 marks**: No relevant content.		6	AO3 3.3.3
	Indicative content (credit any other valid geography, and place-specific detail): • primary data: collected first-hand, so you know the methods and it is specific to your question ; • gives local, up-to-date data of the exact sites you need ; • but is limited by time, small sample, human error and weather ; • secondary data (census, government data, maps, GIS): large samples, often more reliable and covers whole country ; • but may be out of date or not specific, and you do not know how it was collected ; • judgement: both are useful – combining them gives a better enquiry ;			

Where marks are lost in Topic 7

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Paper 3 resources:** always quote evidence (resource number and figure) and use your own knowledge. Marks are for judgement and supported conclusions, not for the option you choose.
- ☐ **Issue evaluation (9 marks):** consider each option's costs, benefits, who it helps and harms, and short vs long term, then give a clear recommendation with a justification.
- ☐ **Maps:** compass directions, scale measurement (use the scale bar and check the units), 4-/6-figure grid references and map symbols can all be tested.
- ☐ **Calculations:** show your working. Mean = total ÷ number; median = middle value (average the two middle ones if even); range = highest − lowest.
- ☐ **Spearman's rank:** rank both sets, find d, square it, add it up, then apply the formula. An answer between −1 and +1 is expected: closer to ±1 = stronger.
- ☐ **Fieldwork methods:** for every method give *what* you measured, *how* (equipment), *where* (sampling) and *why* (how it helps answer the question).
- ☐ **Evaluation of fieldwork:** talk about reliability (repeats), validity (does the data answer the question?), sample size, bias, and how you would improve it.
- ☐ **Presenting data:** match the graph to the data: scatter for relationships, bar for categories, line for change over time, proportional symbols on maps.